ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Manufacturing Industries (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
2	**(d)** Durgapur - Gujarat	Durgapur is in West Bengal
3	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
4	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
5	**(a)** information technology	Software services
6	**(c)** coking coal	Used in blast furnaces
7	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
8	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
9	**(a)** Both A and R are true and R is the correct explanation of A.	These functions of the sector are why it is called the backbone; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R is not the reason for the 'basic' label.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water is scarce (1); Recycling cuts costs and pollution (1).	1+1
12	Solid waste and chemicals contaminate soil (1); Fertility falls and land becomes unfit (1).	1+1
13	Liberalisation opened markets (1); It raised competition and investment (1).	1+1
14	Treatment plants and filters are costly (1); Firms need to invest, but society gains (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The Bhopal tragedy showed safety lapses (1); Rules and training are needed (1); Accidents cost lives (1).	1+1+1
16	Synthetic competition (1); Old machines (1); Falling demand (1).	1+1+1
17	Ports allow import of raw materials (1); They enable exports (1); They attract industries (1).	1+1+1
	OR: Cooperatives pool farmers' resources (1); They assure markets and incomes (1); They develop rural areas (1).	1+1+1
18	Odisha and Jharkhand have bauxite (1); They have cheap power from coal (1); Smelters use both (1).	1+1+1
	OR: Industry creates growth and jobs (1); It also pollutes (1); Balance needs controls (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Thermal pollution (1). (ii) Fly ash or slag (1). (iii) Machinery, generators or drilling (any one) (1). (iv) Treat effluents before discharge and recycle water; cool hot water before release; use rainwater harvesting for processing (any two) (2).	1+1+1+2
20	IT and electronics include software, computers and telecom equipment, and Bengaluru is the electronic capital of India (1); Software Technology Parks provide infrastructure to boost exports and create jobs for educated youth (1); These industries cause little pollution (1); Chemical industries produce fertilisers, pesticides, paints, soaps and drugs and are located in Gujarat, Tamil Nadu, Maharashtra and elsewhere (1); Chemicals support agriculture and industry but also cause pollution (1).	1+1+1+1+1

	OR: Industries are classified by raw material (agro-based and mineral-based), role (basic and consumer), capital (small and large scale), ownership (public, private, joint, cooperative) and bulk of raw material and product (heavy and light) (1); Location depends on availability of raw material, power, labour, capital, transport, markets and government policy (1); Weight-losing raw materials pull industries towards sources: iron and steel near iron ore and coal, sugar near cane fields (1); Cotton textiles grew near cotton and ports in Mumbai and Gujarat, and aluminium near cheap power (1); Industries also cluster near cities and ports for banking, transport and markets, called agglomeration economies (1).	1+1+1+1+1

Mistakes to avoid:

- Placing **Jamshedpur** in Odisha or West Bengal; TISCO is located in **Jharkhand**.
- Mixing classifications: industries by **raw material** (agro-based, mineral-based), by **ownership** (public, private, joint, cooperative) and by **role** (basic, consumer).
- Giving the wrong dates: first successful cotton mill in **Mumbai in 1854**; first jute mill in **1855** at Rishra near Kolkata.
- Saying sugar mills are shifting north; they are moving to **south and west** because of higher sucrose content and a longer crushing season.
- Listing only air pollution as industrial pollution; also note **water** (thermal, effluents), **noise** and **land** pollution.
- Confusing public-sector steel plants (**SAIL**: Bhilai, Durgapur, Rourkela, Bokaro) with the private-sector TISCO.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Manufacturing Industries (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** cooperative industry	Farmers own it
2	**(d)** old technology	Low labour productivity too
3	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
4	**(c)** Visakhapatnam	Located on the coast
5	**(c)** chemicals	Effluents and gases
6	**(c)** liberalisation	Opened economy
7	**(a)** cluster there	Location advantage
8	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
9	**(c)** A is true but R is false.	A is true. Cane in the south and Maharashtra has higher sucrose content, so R is false.
10	**(d)** A is false but R is true.	Industries cause air, water, noise and land pollution, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cars need steel, rubber and glass (1); Growth in autos raises demand for them (1).	1+1
12	Goods like cars need nearness to consumers (1); Market location cuts delivery cost (1).	1+1
13	They attract investment and exports (1); They offer infrastructure and incentives (1).	1+1
14	Farmers own the mills (1); Local control and shared profit raise efficiency (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar mills moved to south and west (1); Cane there has higher sugar content (1); Cooler climate gives a longer crushing season and cooperatives are strong (1).	1+1+1
16	Industries create jobs (1); They raise incomes (1); They reduce poverty (1).	1+1+1
17	Technology raises output (1); It cuts waste (1); It improves quality (1).	1+1+1
	OR: Some regions have more industry (1); Others lack infrastructure (1); Policy should spread growth (1).	1+1+1
18	Cement is used for roads and dams (1); It supports housing (1); Infrastructure growth depends on it (1).	1+1+1
	OR: IT parks provide infrastructure (1); Exports earned foreign exchange (1); They created jobs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Other industries use its products as raw material for machines and tools (1). (ii) Any two of iron ore, coking coal, limestone, manganese (1). (iii) Jharkhand (1). (iv) Iron ore and coal are bulky and heavy and lose weight in processing, so transport cost is reduced by locating near them (2).	1+1+1+2
20	Manufacturing creates jobs, adds to GDP and earns foreign exchange through exports of finished goods (1); It modernises agriculture by supplying fertilisers and machines (1); It reduces regional disparities by locating industries in backward areas (1); Its share in GDP is about 17 per cent, and the National Manufacturing Competitiveness Council was set up in 2006 to raise it to 25 per cent (1); Challenges are power shortages, old technology, low productivity, poor infrastructure and pollution (1).	1+1+1+1+1
	OR: Industries are concentrated in a few regions near cities, ports and raw materials, such as Mumbai-Pune and the Hugli belt (1); Backward areas lack infrastructure, power and skilled labour, and attract few industries (1); Mineral-rich states supply raw materials but have little processing industry (1); Government incentives have not fully balanced growth (1); Better infrastructure and policy support are needed to spread industry (1).	1+1+1+1+1

Mistakes to avoid:

- Placing **Jamshedpur** in Odisha or West Bengal; TISCO is located in **Jharkhand**.
- Mixing classifications: industries by **raw material** (agro-based, mineral-based), by **ownership** (public, private, joint, cooperative) and by **role** (basic, consumer).
- Giving the wrong dates: first successful cotton mill in **Mumbai in 1854**; first jute mill in **1855** at Rishra near Kolkata.
- Saying sugar mills are shifting north; they are moving to **south and west** because of higher sucrose content and a longer crushing season.
- Listing only air pollution as industrial pollution; also note **water** (thermal, effluents), **noise** and **land** pollution.
- Confusing public-sector steel plants (**SAIL**: Bhilai, Durgapur, Rourkela, Bokaro) with the private-sector TISCO.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Manufacturing Industries (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** A is false but R is true.	A false; R true
2	**(b)** infrastructure, labour and markets	Agglomeration economies
3	**(c)** Mumbai in 1854	The Bombay Spinning and Weaving Company
4	**(a)** aluminium smelting	Needs a lot of electricity
5	**(a)** creates demand for steel, glass and rubber	Linked industries
6	**(c)** treating hot water in cooling ponds before release	Cools water first
7	**(a)** lack of safety measures	Poor safety
8	**(d)** A is false but R is true.	A false; R true
9	**(a)** Both A and R are true and R is the correct explanation of A.	These factors favour the mills' location; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are separate facts and R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They employ many people (1); They supply local needs with low capital (1).	1+1
12	Gas is a raw material and fuel for ammonia (1); Fertiliser plants are near gas sources (1).	1+1
13	Old technology and overstaffing raise costs (1); Management and funds problems reduce output (1).	1+1
14	Bulky raw materials are costly to transport (1); Locating near them cuts cost (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rules limit pollution (1); They raise costs for industry (1); They protect health and environment (1).	1+1+1
16	Mills produce yarn and cloth in bulk (1); Handlooms make special cloth (1); Decentralised weaving gives jobs (1).	1+1+1
17	Fertilisers raise yield (1); They support food security (1); Overuse harms soil (1).	1+1+1
	OR: Autos create jobs (1); They add to air pollution (1); CNG and cleaner fuel reduce it (1).	1+1+1
18	Public sector plants provide basic steel (1); Private sector like TISCO is efficient (1); Both are needed but costs are high (1).	1+1+1
	OR: Cleaner technology cuts pollution (1); It saves resources (1); It avoids disasters (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sugarcane is available only in a particular season (1). (ii) Maharashtra (1). (iii) Sucrose content (1). (iv) Cane is bulky and loses its sucrose content if transported far or kept long, so mills near the fields reduce losses and costs (2).	1+1+1+2
20	Agro-based industries use farm raw materials, as in cotton, jute, silk, woollen textiles, sugar and edible oil (1); Mineral-based industries use ores, as in iron and steel, cement, aluminium and petrochemicals (1); Agro-based industries depend on the farm output and are seasonal, like sugar (1); Mineral-based industries depend on mines and are mostly heavy industries (1); Both need power, transport and markets (1).	1+1+1+1+1
	OR: Traditional industries like cotton and jute have declined in relative importance due to old technology and competition (1); Modern industries like IT, electronics and automobiles have grown with new technology and skilled labour (1); Services and manufacturing have become more linked, as IT supports industry (1); Production needs fewer workers per unit and more skills (1); Policy has encouraged modernisation and exports (1).	1+1+1+1+1

Mistakes to avoid:

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MASTERMIND